Summer 2025 Indepedent Study – Bandit Alignment Scratch

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Let $\epsilon = \frac{|\mathcal{A}|^{1/3}}{T^{1/3}}$

$$\begin{aligned}
\mathfrak{R}(\pi, T) &\leq \sqrt{\frac{|\mathcal{A}|T(1-\epsilon)^2}{\epsilon} + 32\epsilon^2 T^2} \cdot \sqrt{|\mathcal{A}| \ln 4T} \\
&\leq \sqrt{\frac{|\mathcal{A}|T}{\epsilon} + 32\epsilon^2 T^2} \cdot \sqrt{|\mathcal{A}| \ln 4T} \\
&= \sqrt{\frac{|\mathcal{A}|T}{\frac{|\mathcal{A}|^{1/3}}{T^{1/3}}} + 32\left(\frac{|\mathcal{A}|^{1/3}}{T^{1/3}}\right)^2 T^2} \cdot \sqrt{|\mathcal{A}| \ln 4T} \\
&= \sqrt{|\mathcal{A}|^{2/3}T^{4/3} + 32|\mathcal{A}|^{2/3}T^{4/3}} \cdot \sqrt{|\mathcal{A}| \ln 4T} \\
&\leq \sqrt{33|\mathcal{A}|^{2/3}T^{4/3}} \cdot \sqrt{|\mathcal{A}| \ln 4T} \\
&\leq \sqrt{33 \ln(4T)}|\mathcal{A}|^{5/6}T^{2/3}
\end{aligned}$$

Compare this to the result in the paper:

$$\mathfrak{R}(\pi, T) \leq \sqrt{33 \ln(4T)}|\mathcal{A}|^{3/4}T^{3/4}$$

Improved scaling with respect to the time horizon T, and worsened with respect to $\mathcal{A}$, the size of the action set. I think this is overall an improvement since we care more about T...

Bibliography